ICAT3110-3011

Intelligent Robots

Project Instructions

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Description

- The submissions will be: **1)** A report **2)** Programming files **3)** A video presentation
- You are required to create a group, then select your **Task (A)** and **Task (B)** from the below set of tasks.
- Based on Task-A, the following attributes should be clearly designed, **<u>mentioning the reason behind</u>**:
    - **<u>Locomotion:</u>**
        - Whether it shall be Wheeled or Legged. And Why?
        - Type and Number of required Wheels or Legs.
        - State the suitable Arrangement of the locomoted parts.
        - Detailed challenges that may face such design regarding Static/Dynamic Stability.
        - State whether your design requires a Suspension system or not.

    - **<u>Kinematic model:</u>**
        - Size and Geometry of the Robot chassis. Why?
        - What should be the material of the Robot body chassis.
        - Proposed approximated net Weight of the chassis (including Wheels/Legs).
        - Stating the required DOF for all End effector parts of the Robot limps.
        - Type and number of motion Actuators.
        - Aided with numerical example, Discuss the degree of Maneuverability.
        - Discuss the Kinematic constraints that should be present in your design.
        - What suits your design best, Holonomic or Non-holonomic Robot drive?
        - State your preferred type of Kinematic control.

    - **<u>Perception:</u>**
        - State what are your desired quantities to be measured as Robot perception.
        - Discuss How many PC/EC and P/A sensors required for the design including -if any- human sensation abilities: Hearing, seeing, feeling and talking. Tabulate your needs, optimize them as much as possible per cost/benefit efficiency.
        - Using Sensors data sheets, describe the evaluation factors of such sensor and their suitability for your design i.e. Accuracy, Sensitivity, Resolution and Precision.
        - Assume what are the possible sources of error -if happens- to your Robot readings.
        - How do you prefer to localize your Robot in the environment? i.e. Using GPS, Fluxgate compass, Ranging Sensors …etc and Why? **<u>Hint</u>**: Consider power limitations for your design.
        - In case of error emergency, What should be the communication channel between the Robot and the human supervisor to correct the robot's failure? Include it in your design.

    - **<u>Localization and Navigation:</u>**
        - Discuss whether your design requires behavior-based or map-based navigation.
        - Design the appropriate positioning and navigation system for your report.
        - Discuss all technical details e.g.: ranging techniques, positioning algorithms, etc
        - Discuss types of navigation methodologies required for your robot in its environment.

- **Draw** a **conceptual drawing** for your robot in the given environment, showing all the above aspects.
- **Finalize** these design requirements into a **<u>full report</u>** and submit it by the due date on Moodle.

Groups (Task A)

- Groups should contain at least **2** members and maximum **6** members.
- Design an Intelligent Robot to perform the required Task A, from the below table of Tasks A.
- No Exceptions to work individually.
- Select one topic from the below list as your **Task (A)/Group No.** , mark your choice in Moodle via this link.

	Application	Environment
Group A1	Mine Sweeper/Detector Robot	Desert
Group A2	Products Organizer Robot	Factory Warehouse
Group A3	Tree Cutter and Grass Trimmer Robot	Rocky Forests
Group A4	Entertainment Robot (play, sing, pet, etc.)	Indoor Buildings
Group A5	Autonomous Trash Collector Robot	Public Parks
Group A6	Pipeline Inspection and Repair Robot	Underground Tunnels
Group A7	Search and Rescue Robot	Collapsed Buildings
Group A8	Precision Farming Robot (planting, spraying)	Agricultural Fields
Group A9	Marine Debris Collection Robot	Coastal Waters
Group A10	Warehouse Inventory Drone	Large Storage Facilities
Group A11	Firefighting and Hazard Detection Robot	Industrial Plants
Group A12	Archaeological Exploration Robot	Deserted Ruins
Group A13	Courier Delivery Robot	Urban Streets
Group A14	Autonomous Window-Cleaning Robot	Skyscrapers
Group A15	Medical Supply Delivery Robot	Hospitals / Clinics
Group A16	Wildlife Monitoring and Tracking Robot	Jungles/Forests
Group A17	Underwater Mapping and Sampling Robot	Deep-Sea Environments
Group A18	Avalanche Victim Detection Robot	Snowy High Mountains
Group A19	Construction Site Inspection Robot	Rough-surface Construction Sites
Group A20	Solar Panel Cleaning and Maintenance Robot	Large-scale Farms / fields
Group A21	Lunar Rover for Regolith Sampling	Moon (Low-Gravity, Dusty Terrain)
Group A22	Martian Cave Explorer Robot	Mars (Subsurface Lava Tubes, Rough Terrain)

Task (B)

- This Task (B) is not necessarily related to your selected Task (A).
- Tasks from DONE Robotics Oy marked as "Company Task", contact them for more info and datasets.
- Select one topic from the below list as your **Task (B)**, mark your choice in Moodle via this link.
- Multiple groups are allowed to choose the same Task (B) simultaneously, up to 3 groups maximum.

	Description	Datasets?
Task B1	**Company task:** Door state tracking and person mapping using computer vision. The robot needs to know the phase of a timed door and the state of a door that stays open/shut. Map all locations of detected persons near the robot. Stretch goal-1: works with swinging, sliding, and elevator doors. Stretch goal-2: Kalman filter for estimating their direction of travel.	Contact DONE Robotics Oy
Task B2	Achieve accurate positioning results for an indoor mobile robot working in a dense environment as Technobothnia lab. You can use any kind of algorithms you think will be effective. Also, you can apply sensor fusion techniques with IMU data.	**Dataset file:** task2_data.zip **Given data:** UWB raw data, IMU data, Robot ground truth data, and illustrative drawings.
Task B3	**Company task:** Voice control interface and follow-me mode. When a user talks with the robot, it should indicate when the user is close enough to activate the mic, and whether or not they are authenticated. Use limited affordances with unauthenticated users. Track a single person and follow him/her without confusing other persons walking by.	Contact DONE Robotics Oy
Task B4	Achieve accurate positioning results for an outdoor mobile robot driving on high-way roads. You can use any kind of algorithms you think will be effective. Also, you can apply sensor fusion techniques with IMU data.	**Dataset file:** task4_data.zip **Given data:** Smart phone raw GNSS data, IMU data, and ground truth data.
Task B5	**Company task:** Detachable control unit / pendant design and implementation: Plan a detachable control unit / pendant for controlling the robot while walking next to it, walking behind it, or standing on it. Includes joystick, enable button, mode switch buttons, status indication, and e-stop button. Charges onboard battery when attached to the robot (magnetically?). Works on the robot or wirelessly. Stretch goal: to actually build one.	Contact DONE Robotics Oy
Task B6	Compare between GNSS and LEO outdoor positioning. Try to improve LEO results with a suitable solution.	**Dataset file:** task6_data.zip **Given data:** GNSS, LEO, IMU data, reference.
Task B7	**Company task:** 3D lidar mapping and robot attachment design: Build a handheld 3D lidar device for mapping a building accurately in 3D. Draft mechanical design alternatives for robot to hospital trolley attachment. Stretch goal: build and test one.	Contact DONE Robotics Oy

Useful Coding Hints for MATLAB/Python

- For downloading "Robotics Toolbox" freely from MathWorks™ website: Click here
- Installation and Adding to MATLAB: Click here
- Getting started with Robotics Toolbox: Click here
- Tutorial "Collision Avoidance" -you can discard importing the CAD design part- : http://bit.ly/2nRwUmj
- Tutorial "Path Planning" -you can discard the CAD design part- : http://bit.ly/2omUMCp
- Adding Path Planning Algorithms on an uploaded map: http://bit.ly/2omUPOB
- The above-mentioned tutorials are just for illustration i.e. not mandatory, you can use your own style.
- All steps are graded, do not worry, you do not have to design a 'Perfect' simulation with zero error(s).
- Do not forget to write '`//comments`' or '`#comments`' beside each step in the m file or the Python file.

Reporting Remarks

- No copying neither from one group to another, nor from online or offline sources.
- You can seek help or illustration from online sources only but not any other individual(s), just self-learning.
- Every claim or piece of information should have a reference, list all your references at the end.
- I expect that you won't use AI tools to do the project on your behalf, this is a fine chance to learn and gain skills for your own personal career. The use of ChatGPT will be detected and the grade will be eliminated.
- You must use the suggested Report template, you can download it from Moodle and follow it.

Video Presentation Remarks

- Each group should record a video to present their project, maximum length of video is **5 minutes.**
- Presentations are in English language, you can use Zoom software for recording, as described in this Link.
- Each group member should present his/her part in the video, again, every member should talk in the video.
- Upload your project report + coding files + video presentation by deadline **17th April 2026** at 23:50.

Submission Notes

- Report Maximum of 15 pages. No copying from any source whatsoever. Write references section at the end.
- Report should be cooperative between all group members **i.e. each member should write his/her name beside the part taken responsibility for e.g.** *Locomotion Part By XXXX, Kinematics Part by YYYY* ,, etc
- Deadline for Report Submission along with coding files and the video is **17th April 2026** at 23:50 on Moodle.